

Deepak Mallampati

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Kent, OH | F-1 OPT, open to sponsorship

PROFESSIONAL SUMMARY

Applied AI engineer (2.5+ years) shipping agentic LLM, RAG, and multi-provider AI products end-to-end - from production healthcare workflows to the Chrome Web Store. Comfortable owning the customer-facing technical lifecycle: deploying GenAI into real engineering workflows, debugging unfamiliar systems, and translating ambiguous problems into deployable AI. Built and shipped a 4-provider multimodal Chrome extension solo (Manga Lens, Chrome Web Store), production agentic healthcare claims pipelines with schema-validated inter-agent contracts, and predictive ML services in FastAPI/Flask + Docker. Self-directed, high-throughput, comfortable with US travel.

CORE COMPETENCIES

Forward Deployed Engineering • Agentic LLM Workflows • Multi-Provider AI Integration • Production RAG • Customer-Facing Delivery • Python • TypeScript • FastAPI • Docker • Evaluation Frameworks • Prompt Engineering • Schema-validated Tool Use • Healthcare AI

PROFESSIONAL EXPERIENCE

Progress Solutions Inc.

Jul 2025 - Present

Jr. AI/ML Engineer Trainee

USA - Healthcare Technology, AI Consulting

- Built **Retrieval-Augmented Generation (RAG) systems** for clinical knowledge retrieval; recursive semantic chunking + transformer embeddings improved contextual retrieval precision ~**35%**, reduced irrelevant retrieval >**30%**, and pushed retrieval-grounded response alignment >**90%**.
- Developed **agentic LLM workflows** for multi-step eligibility checks, care workflow navigation, and documentation clarification with structured tool use, grounding rules, and evaluation audit trails; improved agent response stability ~**25%**.
- Shipped **predictive ML pipelines** (scikit-learn, XGBoost) for patient no-show prediction and care engagement scoring; recall **15-20%** on high-risk categories with precision >**90%**.
- Packaged ML/LLM inference as **FastAPI/Flask** services, containerized with **Docker**, with structured logging; reduced post-deployment defects ~**30%**.
- Owned **customer-facing stakeholder docs**: system-limitation notes, evaluation audit trails, de-identification lineage - closed the loop between deployed AI and clinical end users.

Energy Solutions International (Emerson)

Jun 2022 - Apr 2023

Database & DevOps Performance Engineer (Intern)

Hyderabad, India - Oil & Gas Enterprise ERP

- Tuned T-SQL stored procedures on the Synthesis Order-to-Cash platform; reduced query execution time ~**20%** and data retrieval latency ~**25%**.
- Built **Jenkins CI/CD pipelines** for SQL schema and stored procedure deployments; reduced deployment errors >**30%** and release cycle time ~**35-40%**.
- Built SQL Server DMV + Grafana dashboards for long-running queries, deadlocks, and CPU/I/O contention; cut incident recurrence ~**25%**.

Suvidha Foundation

Jan 2022 - Mar 2022

Machine Learning Engineer

Hyderabad, India - Nonprofit Tech, Applied NLP

- Built a **transformer-based video summarization and highlight system** across 5,000+ sessions; **60-70%** manual review-time reduction with ~**85%** alignment vs human curation.
- Implemented **document Q&A with RAG**; reduced hallucinated outputs ~**30%**; deployed as **Flask API** with a lightweight UI for non-technical staff.

PROJECTS

Manga Lens - Chrome Web Store Extension for Real-Time AI Manga Translation shipped

Shipped Chrome Web Store extension integrating **four AI vision providers** (Claude Sonnet, GPT-4o mini, GPT-4.1 Nano, Ollama/minicpm-v local). Manifest V3, content scripts, service workers, `captureVisibleTab` for panel capture. Tall-webtoon viewport-slice screenshots with coordinate remapping and dedup. Per-domain selector configs for 29 sites, 7-day translation cache, narrowed host permissions for privacy, provider-aware payload routing (WebP for cloud, JPEG for Ollama to avoid CUDA crash).

TypeScript · Manifest V3 · Multi-Provider AI Orchestration

Agentic Healthcare Claims Processing & Fraud Risk Intelligence

Multi-agent pipeline (Intake Normalization → Validation → Consistency → Duplicate Detection → Fraud Risk Scoring) with **schema-validated JSON contracts between agents** to prevent cascading hallucinations. RAG-grounded CPT/ICD validation against vector-indexed policy docs; approximate-nearest-neighbor similarity for duplicate detection; explainable risk scoring with audit-ready reasoning traces.

Python · FastAPI · Vector Search · Structured LLM Tool Use

Agentic Pixel Character Synthesis & Animation Engine

Phased generative AI system for identity-consistent, pose-controlled pixel character synthesis with sprite-sheet export. Stable Diffusion + LoRA for identity, ControlNet + OpenPose/MediaPipe for pose conditioning, and an **LLM-based agent orchestrator** that decomposes high-level prompts into generation tasks.

PyTorch · Diffusers · FastAPI · Docker

Dream Decoder - Multimodal Creative Intelligence Platform

FastAPI + React/TS/Vite/Tailwind app coordinating Perplexity Sonar (interpretation) and Replicate (imagery). Intermediate **structured prompt transformation layer** improved contextual alignment **~30%** and first-pass image success **~25-30%** over naïve prompting.

FastAPI · React · Multimodal API Orchestration

Driver Drowsiness Detection - Real-Time YOLOv8

Replaced two-stage CNN with unified **YOLOv8** detection-and-classification; reduced inference latency **~30%**. Sliding-window confidence aggregation suppressed blink false positives **~25%**.

PyTorch · OpenCV · YOLOv8

EDUCATION

Master's Degree, Kent State University, Ohio, USA

Aug 2023 - May 2025

CERTIFICATIONS

- see Skills & Projects for portfolio depth

SKILLS

Languages & Frameworks: Python, TypeScript, FastAPI, Flask, React, SQL (PostgreSQL, T-SQL, SQLite), C++ (Arduino), PHP

LLM & GenAI: Agentic workflows, RAG, structured outputs, prompt engineering, evaluation pipelines, guardrails, grounding, semantic retrieval, embeddings, vector search, multi-provider orchestration (Claude, GPT, Ollama)

ML Libraries: scikit-learn, XGBoost, PyTorch, Hugging Face Transformers, Diffusers, LangChain, LlamaIndex

Computer Vision & Multimodal: YOLOv8, OpenCV, ControlNet, OpenPose/MediaPipe, Stable Diffusion

Infra & DevOps: Docker, Jenkins, Grafana, CI/CD, RESTful APIs, observability, structured logging

Domains: Applied AI, Forward Deployed Engineering, Healthcare AI (HIPAA-conscious), Multi-Modal AI, Workflow Automation